Universität Bonn

Notes for the lecture

Combinatorics, Graphs, Matroids

held by

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TEXed by

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Winter 2024/25

Corrections and improvements

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Lecture

Part I

Combinatorics

Chapter 1

Introduction

1.1 Fundamental Counting Rules

[08.10.2024, Lecture 1]

Goal: Count the number of elements in some finite sets.

Lemma 1.1: Fundamental counting rules

(*i*) *For pairwise disjoint and finite sets* $U_1, \dots, U_n$*, we have*

$$\left| \bigcup_{i=1}^{n} U_i \right| = \sum_{i=1}^{n} |U_i| . \tag{1.1}$$

(*ii*) *For finite sets* $U_1, \dots, U_n$*, we have*

$$|U_1 \times \cdots \times U_n| = \prod_{i=1}^{n} |U_i| . \tag{1.2}$$

(*iii*) *If there is a bijection between two sets* S *and* T*, then* $|S| = |T|$*.*

Notation. We call a set a n-set if it is a set with cardinality $n \in \mathbb{N}$ (where $0 \in \mathbb{N}$).

Proposition 1.2. *For two* n*-sets* A *and* B*, the number of bijections between* A *and* B *is* $n!$*.*

Proof. Let U the set of bijections from A to B. Apply induction in n:

$n = 0$: Trivial, because $0! = 1 = |U|$.

$n \mapsto n + 1$: Let $n > 0$ and let $x \in A$. For $i \in B$, let U_i be the set of all bijections $f : A \to B$ with $f(x) = i$. By induction, $|U_i| = (n-1)!$ for each $i \in B$. Then, $U = \bigcup_{i \in B} U_i$, so we get

$$|U| = \sum_{i \in B} |U_i| = n \cdot (n-1)! = n!. \tag{1.3}$$

□

Corollary 1.3. *The number of permutations of an* n*-set is* $n!$*.* □

Proposition 1.4. *The number of mappings from a* k*-set* A *to an* n*-set* B *is* n^k*.*

Proof. There is a bijection between the set of all mappings from A to B and the set $\underbrace{B \times \cdots \times B}_{k \text{ times}}$.

The number of these mappings is by Lemma 1.1.(*ii*) just $|B|^k = n^k$. □

Notation. We denote the set of all mappings from A to B by B^A.

Proposition 1.5. *The number of subsets of an n-set is 2^n.*

Proof. For an n-set A, there is a bijection between its power set and the mappings from A to $\{0,1\}$. Thus, the power set has 2^n elements. □

Notation.

(*i*) We denote the power set of a set A by 2^A.

(*ii*) The *binomial coefficient* $\binom{n}{k}$ is defined as the number of k-subsets of a n-set (for $k, n \in \mathbb{N}$).

What does *counting* mean? $\rightsquigarrow$ Possible answers:

(*i*) a direct, closed formula,

(*ii*) a sum,

(*iii*) a recursive formula

Example. Consider the number y_n of $0-1-2$-strings of length n with an even number of 1s and an odd number of 2s. Answers of types (*ii*) and (*iii*):

- We have
$$y_n = \sum_{i=0}^{\lfloor \frac{n-1}{2} \rfloor} \binom{2}{2i+1} 2^{2i}. \tag{1.4}$$
- We have
$$y_n = 3^{n-1} - y_{n-1} \quad (n > 0). \tag{1.5}$$

The proof of both formulas is exercise 1.2.

1.2 Elementary Counting Coefficients

Proposition 1.6. *For $n, k \in \mathbb{N}$ with $k \le n$, we have*

(*a*) $\binom{n}{k} = \binom{n}{n-k}$

(*b*) $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$ *for* $k \ge 1$

(*c*) $\binom{n}{k} = \frac{n!}{(n-k)!k!}$.

Consider three kinds of rows in Pascal's triangle:

Row sums $\sum_{k=0}^{n} \binom{n}{k} = 2^n$ (both terms count the number of subsets of an n-set)

Column sums $\sum_{m=0}^{n} \binom{m}{k} = \binom{n+1}{k+1}$ for $n, k \in \mathbb{N}$ (where $\binom{m}{k}$ is the number of all $(k+1)$-subsets of $\{1, \ldots, n+1\}$ with the property that $m+1$ is the largest element in the subset)

Diagonal sums $\sum_{k=0}^{n} \binom{m+k}{k} = \binom{m+n+1}{n}$ for $m, n \in \mathbb{N}$ (for $k \in \{0, \ldots, n\}$, where $\binom{m+n}{k}$ is the number of n-subsets of $\{1, \ldots, m+n+1\}$ where $m+k+1$ is the largest number that is **not** contained in the subset)

Proposition 1.7. *For $x, y \in \mathbb{R}$, we have $(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}$.*

Proof. The coefficients of $x^k y^{n-k}$ in $(x+y)^n$ is $\binom{n}{k}$. □

Definition 1.8: $S_{n,k}$: Stirling number of the second kind

For $n, k \in \mathbb{N}$ let $S_{n,k}$ be the number of ways to partition an n-set in to k non-empty sets. The number $S_{n,k}$ is called *Stirling number of the second kind.*

Observation. For $n > 0$: $S_{n,0} = 0, S_{n,1} = n, S_{n,2} = 2^{n-1} - 1$.

Proposition 1.9. *For $n, k \in \mathbb{N} \setminus \{0\}$ we have $S_{n,k} = S_{n-1,k-1} + k \cdot S_{n-1,k}$.*

Proof. Let M be an n-set and $x \in M$. Then, the set of partitions of n into k subsets can be decomposed in the set of partitions with the set $\{x\}$ (there are $S_{n-1,k-1}$ of them) and the set of partitions where x is an element in a set with more than one element (there are $k \cdot S_{n-1,k}$ of them) □

[08.10.2024, Lecture 1]
[10.10.2024, Lecture 2]

Next goal: Count the number of mappings from an n-set to an r-set under certain constraints. To this end, we collect *counting coefficients*. We know already:

- $\binom{n}{k}$
- $S_{n,k}$: Number of ways to partition an n-set into k non-empty sets

Definition 1.10: $P_{n,k}$: Number of ways writing n as sum of k integers

For $n, k \in \mathbb{N}$, let $P_{n,k}$ be the number of ways to write n as the sum of k positive integers (without considering the order of the summands).

Example. $P_{6,3} = 3$, because the only possible partitions are: $6 = 4+1+1 = 3+2+1 = 2+2+2$.

Observation. For $n > 0$, we have $P_{n,0} = 0, P_{n,1} = 1, P_{n,2} = \lfloor \frac{n}{2} \rfloor$.

We can also consider *ordered partitions of numbers*:

Example. There are 6 ordered 3-partitions of the number 5: $5 = 3+1+1 = 1+3+1 = 1+1+3 = 2+2+1 = 2+1+2 = 1+2+2$.

Proposition 1.11. *The number of ordered partitions of a positive integer n into k summands is $\binom{n-1}{k-1}$.*

Proof. Write n as $n = \underbrace{1 + \cdots + 1}_{n \text{ times}}$. Any choice of $k-1$ of the $n-1$ „+“-signs gives you an ordered partition of n into k summands. Tis is a one-to-one correspondance. □

1.3 Assignments

Observations.

(*i*) The number of *surjective mappings* from an n-set N to an r-set R is $r! \cdot S_{n,r}$.

Proof. There are $S_{n,r}$ ways to partition n into r preimages and for each such partiton, we get $r!$ ways to assign the preimages to the elements of R. □

(*ii*) The number of *injective mappings* from an n-set N to an r-set R is

$$r(r-1)(r-2)\cdots(r-n+1). \tag{1.6}$$

Notation. For $n, r \in \mathbb{N}$, we call

$$r^{\underline{n}} := r \cdot (r-1) \cdot (r-2) \cdots (r-n+1) \tag{1.7}$$

the *n-th falling factorial* of r.

Analogously, the *n-th rising factorial* of r is

$$r^{\overline{n}} := r \cdot (r+1) \cdot (r+2) \cdots (r+n-1). \tag{1.8}$$

Definition 1.12: Multiset

A *k-multiset* over an n-set A is a set $B \subseteq A \times \mathbb{N}$ such that $\sum_{(a,i)\in B} i = k$ and $|\{(a,i) \in B \mid i \in \mathbb{N}\}| = 1$ for all $a \in A$.

$\rightsquigarrow$ Multisets can be considered as a set that may contain an element more than once. We write again $\{a_1, \ldots, a_k\}$ for a k-multiset (so $a_i = a_j$ is possible for $i \neq j$).

Proposition 1.13. *The number of k-multisets over an n-set is $\frac{n^{\overline{k}}}{k} = \binom{n+k-1}{k}$.*

Proof. There is a bijection f between the set of all k-multisets over $\{1, \ldots, n\}$ and the set of all k-subsets of $\{1, \ldots, n+k-1\}$. For a k-multiset $\{a_1, \ldots, a_k\}$ with $a_1 \leq a_2 \leq \cdots \leq a_{k-1} \leq a_k$, set $f(\{a_1, \ldots, a_j\}) = \{a_1, a_2+1, a_3+2, \ldots, a_k+(k-1)\}$. It is easy to see that this defines a bijection. □

Proposition 1.14. *For $n, r \in \mathbb{N}$, we have $r^n = \sum_{k=0}^{n} S_{n,k} r^{\underline{k}}$.*

Proof. For two sets A and B, let $\mathrm{Surj}(A,B)$ be the set of all surjective mappings from A to B. We know from Observation (i) that $|\mathrm{Surj}(A,B)| = |B|! \cdot S_{|A|,|B|}$. Let N be an n-set and R an r-set. Then, r^n is the number of mappings from N to R. We can partition

$$\begin{aligned} r^n &= \sum_{A \subseteq R} |\mathrm{Surj}(N,A)| = \sum_{k=0}^{r} \sum_{\substack{A \subseteq R \\ |A|=k}} |\mathrm{Surj}(N,A)| \\ &= \sum_{k=0}^{r} \binom{n}{k} k! S_{n,k} = \sum_{k=0}^{r} S_{n,k} r^{\underline{k}} = \sum_{k=0}^{n} S_{n,k} r^{\underline{k}} \end{aligned} \tag{1.9}$$

where the last step is possible because $S_{n,k} = 0$ for $k > n$ and $r^{\underline{k}} = 0$ for $k > r$. □

elements distinguishalbe		Mapping $f : N \to R$			
N	R	arbitrary	injective	surjective	bijective
Yes	Yes	r^n	$r^{\underline{n}}$	$r! S_{n,r}$	$\begin{matrix} 0 & (n \neq r) \\ n! & (n = r) \end{matrix}$
No	Yes	$\frac{r^{\overline{n}}}{n!}$	$\binom{r}{n}$	$\binom{n-1}{r-1}$	$\begin{matrix} 0 & (n \neq r) \\ 1 & (n = r) \end{matrix}$
Yes	No	$\sum_{k=1}^{r} S_{n,k}$	$\begin{matrix} 0 & (r < n) \\ 1 & (r \geq n) \end{matrix}$	$S_{n,r}$	$\begin{matrix} 0 & (n \neq r) \\ 1 & (n = r) \end{matrix}$
Yes	Yes	$\sum_{k=1}^{r} P_{n,k}$	$\begin{matrix} 0 & (r < n) \\ 1 & (r \geq n) \end{matrix}$	$P_{n,r}$	$\begin{matrix} 0 & (n \neq r) \\ 1 & (n = r) \end{matrix}$

Table 1.1: Number of mappings from an n-set N to an r-set R with and without additional constraints

First row completely covered.

Second row
- $\frac{r^{\overline{n}}}{n!}$ is the number of r-multisets of an n-set.
- $\binom{n-1}{r-1}$ is the number of ordered partitions of n into r summands.

1.4 Generalized Counting Coefficients

Definition 1.15: Generalized factorials and coefficients

For $r \in \mathbb{C}$ and $k \in \mathbb{N}$, we define

(*i*) $r^{\underline{n}} := r \cdot (r-1) \cdot (r-2) \cdots (r-k+1)$ (with $r^{\underline{0}} = 1$)

(*ii*) $r^{\overline{n}} := r \cdot (r+1) \cdot (r+2) \cdots (r+k-1)$ (with $r^{\overline{0}} = 1$)

For $r \in \mathbb{C}, k \in \mathbb{Z}$, we define

$$\binom{r}{k} = \begin{cases} \frac{r^{\underline{k}}}{k!} & k \geq 0 \\ 0 & k < 0. \end{cases} \tag{1.10}$$

Proposition 1.16. *For $r \in \mathbb{C}, k \in \mathbb{Z}$, we have*

$$\binom{r}{k} = \binom{r-1}{k-1} + \binom{r-1}{k}. \tag{1.11}$$

Proof. If $k \leq 0$: trivial.

For $k > 0$, both sides of the equation are (for a fixed k) polynomials in r of degree k. They are equal for any $r \in \mathbb{N}$, so for an infinite number of values of r. $\rightsquigarrow$ Both polynomials are identical. □

[10.10.2024, Lecture 2]
[15.10.2024, Lecture 3]

Proposition 1.17 (Vandermonde identity)**.** *For $x, y \in \mathbb{C}$ and $n \in \mathbb{N}$, we have*

$$\binom{x+y}{n} = \sum_{k=0}^{n} \binom{x}{k} \binom{y}{n-k}. \tag{1.12}$$

Proof. Again, we can use the polynomial method: For $x, y \in \mathbb{N}$, this is true because both terms describe the number of k-subsets of an $(x+y)$-set. For general values of x and y, the statement follows because both terms are polynomials of degree n and are identical on an infinite number of points. □

Proposition 1.18. *For $r \in \mathbb{C}$ and $k \in \mathbb{Z}$, we have*

$$\binom{-r}{k} = (-1)^k \binom{r+k-1}{k}. \tag{1.13}$$

Proof. We have

$$(-r)^{\underline{k}} = (-r)(-r-1)\cdots(-r-k+1) = (-1)^k r(r+1)\cdots(r+k-1) = (-1)^k r^{\overline{k}}. \tag{1.14}$$

Dividing this equation by $k!$ proves the claim. □

Remark. Together with the equation $\sum_{k=0}^{n} \binom{m+k}{k} = \binom{m+n+1}{n}$ (see the observation about diagonal sums on Pascal's triangle) that is valid for any $m \in \mathbb{C}$, we get

$$\sum_{k=0}^{m} (-1)^k \binom{n}{k} = \sum_{k=0}^{m} \binom{k-n-1}{k} = \binom{m-n}{m} = (-1)^m \binom{n-1}{m}. \tag{1.15}$$

Hence, we have a formula for the alternating sum of a row in Pascal's triangle. Moreover, the equation has a combinatorical interpretation: Both sides count the number of m-subsets of $\{1,\dots,n-1\}$ if m is even (or minus this number if m is odd).

1.5 Permutations

An *n-permutation* is a bijection $\pi : \{1,\dots,n\} \to \{1,\dots,n\}$.

For a permutation π, a *cycle* is a vector $(i_1, i_2, \dots, i_t)$ such that $\pi_(i_j) = i_{j+1}$ for $j \in \{1,\dots,t-1\}$ and $\pi(i_t) = i_1$. Hence, a fixed point correspondents to a cycle of length 1.

Definition 1.19: $s_{n,k}$: Stirling number of the first kind

For $n, k \in \mathbb{N}$ let $s_{n,k}$ be the number of n-permutations with k cycles (where we set $s_{0,0} := 1$). The number $s_{n,k}$ is called *Stirling number of the first kind*.

Obviously, for $n > 0$ we have: $s_{n,0} = 0, s_{n,1} = (n-1)!$. As for the Stirling numbers of the second kind, we get a recursion formula.

Proposition 1.20. *For $n, k \in \mathbb{N} \setminus \{0\}$, we have*

$$s_{n,k} = s_{n-1,k-1} + (n-1)s_{n-1,k}. \tag{1.16}$$

Proof. There are $s_{n-1,k-1}$ n-permutations with k cycles where n is a fixed point and $(n-1)s_{n-1,k}$ n-permutatios with k cycles where n is not a fixed point. □

For the Stirling numbers of the second kind, also the number $Sn, 2$ (for $n > 1$) was easy to compute. To do this for the Stirling numbers of the first kind, we need the harmonic numbers.

Definition 1.21: Harmonic numbers

For $n \in \mathbb{N}$, we define $H_n := \sum_{i=1}^{n} \frac{1}{i}$. We call H_n the *n-th harmonic number*.

It is a well-know fact (that can easily be proved by using the fact that the derivative of $\ln(x)$ is $\frac{1}{x}$) that $H_n \in \Theta(\ln(n))$; more precisely, we have for positive integers n:

$$\ln(n) + \frac{1}{n} \le H_n \le \ln(n) + 1. \tag{1.17}$$

Proposition 1.22. *For $n \in \mathbb{N} \setminus \{0\}$, we have $s_{n,2} = (n-1)!H_{n-1}$.*

Proof. Induction in n. For $n = 1$, we have 0 on both sides. So let $n > 1$. Then:

$$\frac{s_{n,2}}{(n-1)!} = \frac{s_{n-1,1}}{(n-1)!} + \frac{(n-1)s_{n-1,2}}{(n-1)!} = \frac{(n-2)!}{(n-1)!} + \frac{s_{n-1,2}}{(n-2)!} = \frac{1}{n-1} + H_{n-2} = H_{n-1}. \tag{1.18}$$

□

There is a surprising connection between the two kinds of Stirling numbers: They are coefficients that allow to switch between the bases $\{r^0, r^1, r^2, \dots\}$ and $\{r^{\underline{0}}, r^{\underline{1}}, r^{\underline{2}}, \dots\}$ of the vector space of the polynomials. We know already the following equation for $n, r \in \mathbb{N}$:

$$r^n = \sum_{k=0}^{n} S_{n,k} r^{\underline{k}}. \tag{1.19}$$

By the polynomial method, we can generalize this result to $r \in \mathbb{C}$. For the other direction (i.e. for writing $r^{\underline{n}}$ as a linear function of the polynomials r^k), we can use the Stirling numbers of the first kind:

Proposition 1.23. *For $n \in \mathbb{N}$ and $r \in \mathbb{C}$, we have*

$$r^{\underline{n}} = \sum_{k=0}^{n} (-1)^{n-k} s_{n,k} r^k. \tag{1.20}$$

Proof. Induction in n. For $n = 0$, both sides of the equation are equal to 1. Hence, assume $n > 0$. We generalize the definition of $s_{n,k}$ to negative values of k by setting $s_{n,k} := 0$ for $k < 0$ and get:

$$r^{\underline{n}} = (r-n+1)r^{\underline{n-1}} = (r-n+1)\sum_{k=0}^{n-1}(-1)^{n-1-k} s_{n-1,k} r^k \tag{1.21}$$

$$= \sum_{k=0}^{n-1}(-1)^{n-1-k} s_{n-1,k} r^{k+1} + \sum_{k=0}^{n-1}(-1)^{n-k}(n-1) s_{n-1,k} r^k \tag{1.22}$$

$$= \sum_{k=1}^{n}(-1)^{n-k} s_{n-1,k-1} r^k + \sum_{k=0}^{n}(-1)^{n-k}(n-1) s_{n-1,k} r^k \tag{1.23}$$

$$= \sum_{k=0}^{n}(-1)^{n-k}(s_{n-1,k-1} r^k + (n-1) s_{n-1,k} r^k) = \sum_{k=0}^{n}(-1)^{n-k} s_{n,k} r^k \tag{1.24}$$

□

We categorize permutations not only be the number of cycles but also on the number of cycles of a certain length.

Notation. For a permutation π, there let $b_i(\pi)$ be the number of cycles of length i (for $i \in \{1,\dots,n\}$), and let $b(\pi) := \sum_{i=1}^{n} b_i(\pi)$ be the total number of cycles. The *type of a permutation* π is the formal term $t(\pi) = 1^{b_1(\pi)} 2^{b_2(\pi)} \cdots n^{b_n(\pi)}$ where we skip the numbers i with $b_i(\pi) = 0$.

Observation. $\sum_{i=1}^{n} i b_i(\pi) = n$.

Proposition 1.24. *There are $\sum_{k=0}^{n} P_{n,k}$ types of n-permutations.*

Proof. For every type $1^{b_1(\pi)} 2^{b_2(\pi)} \cdots n^{b_n(\pi)}$, we find a representation of n as a sum of $b(\pi)$ positive integers where the number i occurs $b_i(\pi)$ times as a summand. This leads to a bijection between the set of types of an n-permutations and the set of partitions of n. □

Proposition 1.25. *There are*

$$\frac{n!}{b_1! \cdots b_n! 1^{b_1} 2^{b_2} \cdots n^{b_n}} \tag{1.25}$$

n-permutations of type $1^{b_1} 2^{b_2} \cdots n^{b_n}$.

Proof. First consider the empty cycles

$$\underbrace{(\cdot)(\cdot)\dots(\cdot)}_{b_1 \text{ times}}\underbrace{(\cdot\cdot)(\cdot\cdot)\dots(\cdot\cdot)}_{b_2 \text{ times}}\underbrace{(\cdots)(\cdots)\dots(\cdots)}_{b_3 \text{ times}}\dots \tag{1.26}$$

There are $n!$ ways to fill these cycles with numbers from $\{1,\dots,n\}$ (without repeating numbers). Moreover, each such assignment yields an n-permutation of type $1^{b_1} 2^{b_2} \cdots n^{b_n}$. However, for each $i \in \{1,\dots,n\}$, all assignments that differ just in the order of the b_i cycles lead to the same permutation. And for each cycle of length i, there are i assignments giving the same permutation

because there are i ways to choose the first element in the cycle. Hence, for each n-permutation π, there $b_1! \cdots b_n! 1^{b_1} 2^{b_2} \cdots n^{b_n}$ assignments that encode π. □

In particular, we have

$$s_{n,k} = \sum_{(b_1,\ldots,b_n)} \frac{n!}{b_1! \cdots b_n! 1^{b_1} 2^{b_2} \cdots n^{b_n}} \tag{1.27}$$

where we sum over all vectors $(b_1, \ldots, b_n)$ with $\sum_{i=1}^{n} i b_i = n$ and $\sum_{i=1}^{n} b_i = k$.

Similarly:

$$n! = \sum_{(b_1,\ldots,b_n)} \frac{n!}{b_1! \cdots b_n! 1^{b_1} 2^{b_2} \cdots n^{b_n}} \tag{1.28}$$

where we sum over all vectors $(b_1, \ldots, b_n)$ with $\sum_{i=1}^{n} i b_i = n$.

[15.10.2024, Lecture 3]
[17.10.2024, Lecture 4]

Appendix

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